MILITARY SPECIFICATION

FIRE CONTROL COMPONENTS AND ASSEMBLIES

This amendment forms a part of Military Specification MIL-F-45133B(Ord), dated 18 January 1962.

Paragraph 2.1: Add the following Military documents:

"MIL-S-6872 Soldering Process, General Specification For
MIL-S-45743 Soldering, Manual Type, High Reliability, Electrical Connections, For Missile Systems, Procedures For
* MIL-S-46844 Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type
MIL-B-60621 Bonding Optical Assemblies (Glass to Metal)
* MIL-STD-130 Identification Marking of U.S. Military Property"

Paragraph 2.1: Delete "MIL-STD-10" and "title". Add: "MIL-STD-100 Engineering Drawing Practices."

Paragraph 2.2: Add:

"American Standard

ASA B46.1-1962 Surface Texture
ANSI Y14.5-1973 Dimensioning and Tolerancing

(Non-military activities obtain copies of ASA and ANSI Standards from: American National Standards Institute, 1430 Broadway, New York City, New York, 10018.

Paragraph 3.3: Add: "interpretation of tolerance symbols specified on drawings shall be as defined in ANSI Y14.5 - 1973."

Paragraph 3.4: Retain Table I. Delete paragraph and substitute:

"3.4 Machine finish.- Where letter symbols for surface roughness of machined finishes are specified on drawings, roughness height as defined in American Standard ASA B46.1-1962 shall be in accordance with Table I. The specified finish value represents the roughest surface acceptable, regardless of the material specified or the process used to obtain the finish."

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Paragraph 3.7.2: Delete "800 volts" and substitute "500 volts".

* Paragraph 3.8: Delete and substitute:

"3.8 Part identification and marking requirements.- Part identification and marking requirements shall be in accordance with MIL-STD-130, Identification Marking of U.S. Military Property."

Paragraph 3.9.4: Delete and substitute:

"3.9.4 Soldering.

3.9.4.1 Soldering of electrical connections.- Soldering of electrical connections and certification of soldering operators shall be in accordance with MIL-S-45743.

3.9.4.2 Soldering, except electrical connections.- Soldering, other than electrical connections, shall be in accordance with MIL-S-6872. The solder type designation shall be as specified on the applicable drawing."

* Paragraph 3.9.4.3: Add the following:

"3.9.4.3 Soldering of printed wiring assemblies.- Soldering of printed wiring assemblies shall be in accordance with MIL-S-46844, Solder Bath, Soldering of Printed Wiring Assemblies, Automatic Machine Type."

3.9.5: Delete and substitute:

"3.9.5 Bonded joints.- The bonding of optical glass components to their metal supports shall be in accordance with MIL-B-60621. Bonding of other than glass to metal shall be in accordance with the recommended practice of the material manufacturer unless specified techniques are prescribed on the drawings or referenced specifications."

Paragraph 3.10: Add the following:

"3.10 Source control drawings.- A source control drawing is one which defines the required item entirely, or in part, by means of a vendor's part number. It may include additional functional, test or configuration specifications but since these are in themselves inadequate to assure proper performance, installation, and interchangeability of the item in a specific application, procurement of the item is thus effectively restricted to the approved source or sources listed on the drawing.

3.10.1 Substitute items.- Where source control drawings in accordance with the provisions of MIL-STD-100 (Engineering Drawing Practices) are included in the contractual technical data package for commercial or vendor items, the contractor shall supply only the approved item(s) specified on the drawing unless written approval is obtained from the Contracting

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Office for use of a substitute item. The contractor is cautioned, however, that the acquisition of data necessary to substantiate the physical and functional interchangeability of a proposed substitute for an approved source control item, and its long-term reliability in its intended environment, may require facilities, equipment, and resources not available or practically obtainable at the time approval of a substitute item is requested. The Government therefore makes no representation as to the feasability or probability of approving a substitute item. Inasmuch as the contractor is obligated to supply the specified item or an approved equal, rejection or delay in approval by the Government of a proposed substitute item shall not be construed as justification for any change in contract terms and conditions.

3.10.2 Unavailability of specified approved item.- Where an item specified on a source control drawing can be documented by the contractor to have been commercially available to him at the time of submitting a bid or proposal, but should thereafter be withdrawn from the commercial market, it shall be the contractor's responsibility to submit evidence of the interchangeability and reliability of an available substitute item. The Government will in such cases assume responsibility for such evaluation and testing as it deems necessary to assure equality of the proposed substitute; however, unless the Government was advised of the unavailability of the specified item prior to contract finalization, delay in proposing and obtaining approval of the substitute item shall not be cause for change in the terms and conditions of the contract.

3.10.3 Suitability of approved item.- Unless otherwise specified in the contract, the Government will assume responsibility for the suitability of approved items for their intended use, provided that the items comply with all specified requirements of the source control drawing."

Paragraph 4.4.3: Add the following subparagraphs:

"4.4.3.1 Contractor design of test equipment.- The contractor shall design and supply inspection equipment compatible with the component inspection procedures specified in 4.6.6. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.3.2 Gaging for screw threads.- Thread gages utilized for inspection of screw threads shall be made in accordance with the National Bureau of Standards Handbook H-28."

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Paragraph 4.5: Delete and substitute:

"4.5 Category C - Certification.- Where Category C Certification is authorized in the contract, the contractor shall certify that the product being supplied has been determined to be in full compliance with all requirements of the drawings and specifications cited in the contract and that workmanship and materials conform with recognized standards of commercial quality. The certification shall reference the contract or purchase order number, and shall include part number, nomenclature, lot number and quantity for the lot or shipment, analytical results from each test together with stated specification limits, and name of the company. The certificate shall contain a statement as follows:

> As the authorized representative of the contractor, the undersigned warrants and represents that: All of the information supplied herewith is true and accurate; the material and analyses covered by this certificate conforms to all the contract requirements (including but not limited to the drawings and specifications); and this certificate is made for the purpose of inducing payment and with knowledge that the information and certification may be used as a basis for such payment."

Paragraph 4.6.6.1.1: Delete and substitute:

"4.6.6.1.1 Alternate procedure.- Where a SQAP specifies the use of TABLES II and III in MIL-F-45133, MIL-STD-105 shall be utilized in lieu of the deleted TABLES II and III with the following AQL's applicable except where specific AQL's are designated:

> Major Defects - AQL 0.65%
> Minor Defects - AQL 2.5% "

Paragraph 4.6.6.2: Delte and substitute:

"4.6.6.2 Components not covered by SQAP.- Components not covered by SQAP shall be inspected in accordance with sampling plan tables of MIL-STD-105 and the AQL's specified in 4.6.6.1.1."

Page 12, TABLES II and III: Delete in their entirety.

Page 14, paragraph 6.2.2: Add "and one copy to the consignee attached to the packing list."

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* The outside margins have been marked to indicate where changes to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous amendment.

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